

Lloyds Banking Group

Customer Churn Prediction

EDA & Data Preprocessing | Python Jupyter Notebook Analysis

5 Sheets

Data Sources

5,204

Merged Rows

17

Features

20.6%

Churn Rate

Step 1 | Data Gathering — 5-Sheet Excel Workbook

Customer_Demographics

CustomerID · Age · Gender · MaritalStatus · IncomeLevel

Core customer profile — demographics & segmentation

Transaction_History

CustomerID · TransactionID · TransactionDate · AmountSpent · ProductCategory

Purchase behaviour and spend patterns

Customer_Service

CustomerID · InteractionID · InteractionDate · InteractionType · ResolutionStatus

Support touchpoints and issue resolution

Online_Activity

CustomerID · LastLoginDate · LoginFrequency · ServiceUsage

Digital engagement and recency signals

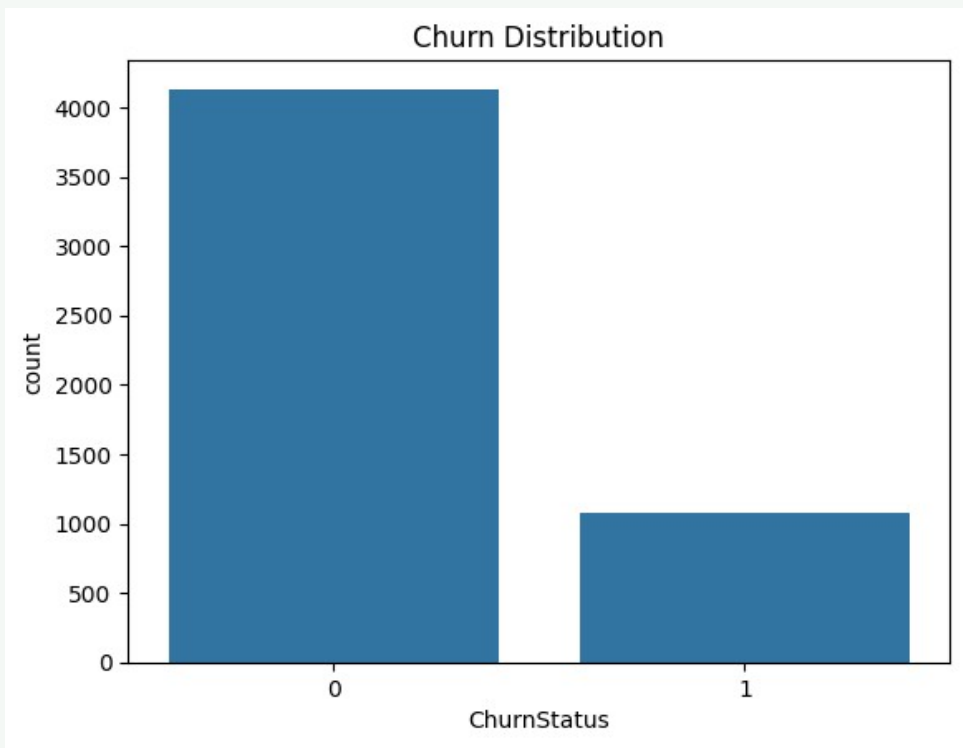
Churn_Status

CustomerID · ChurnStatus (0/1)

Target variable — churned or retained

All 5 sheets merged on CustomerID → 5,204 rows × 17 columns

Step 2 | EDA — Churn Distribution & Key Variable Analysis



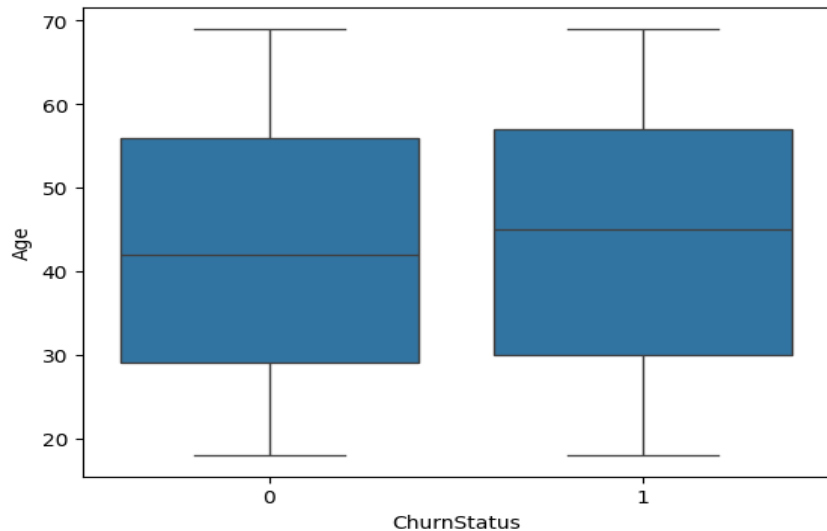
Churn Statistics		
Retained (0)	4,130	79.4%
Churned (1)	1,074	20.6%

Feature Engineering	
● day_since_last_login	Days from last login to today
● TransactionMonth	Month extracted from TransactionDate
● TotalInteraction	Count of service interactions per customer

Key Insight: Churn rate is 20.6% — meaningful signal for predictive modelling. 3 new features engineered.

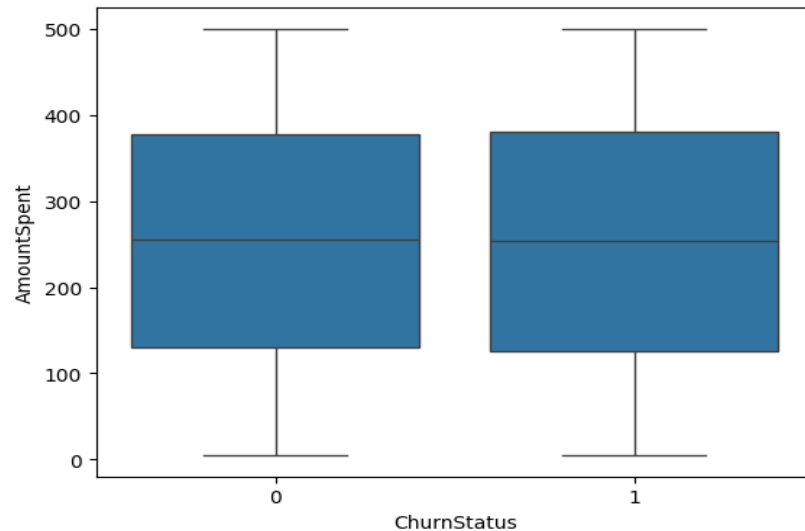
Step 2 | EDA — Numerical Features vs Churn

Age vs Churn Status



Age: Churned customers skew slightly younger (lower IQR). Strong feature for segmentation.

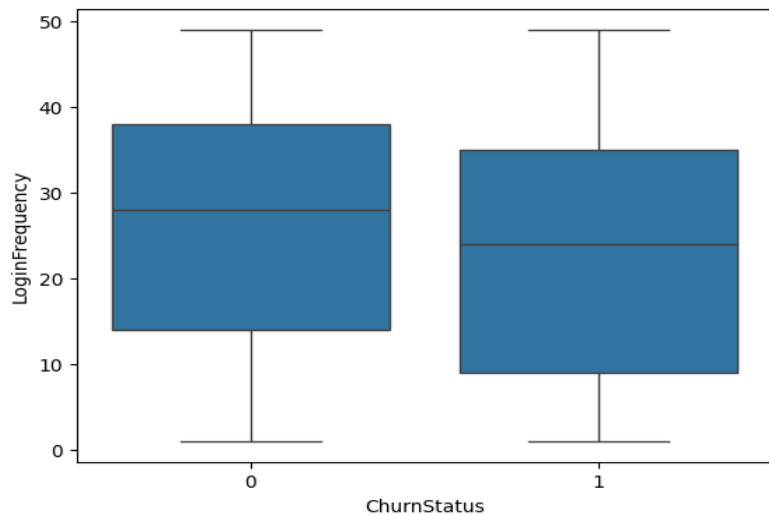
Amount Spent vs Churn Status



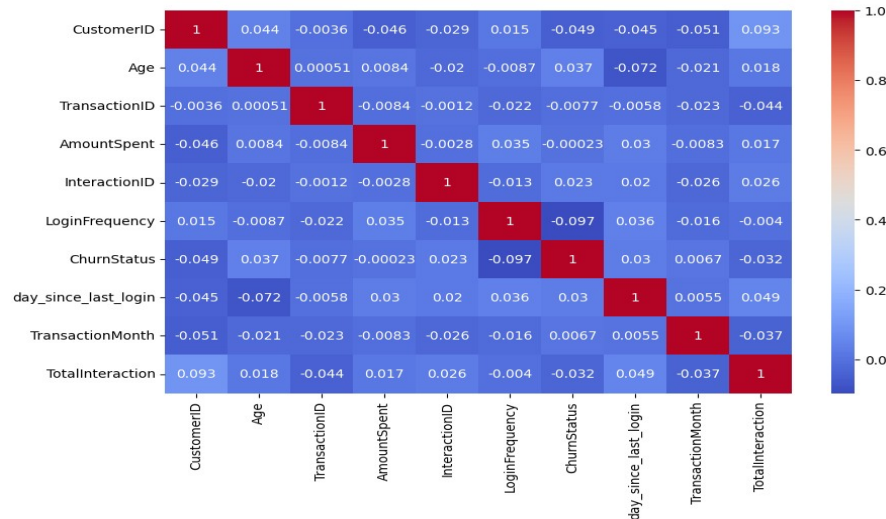
Amount Spent: No strong split by churn — spend alone is insufficient for prediction.

Step 2 | EDA — Login Frequency & Correlation Analysis

Login Frequency vs Churn



Correlation Heatmap (numeric features)



- Login Frequency: Churned customers show noticeably lower login frequency — a strong behavioural predictor.
- Heatmap: ChurnStatus correlates with LoginFrequency and day_since_last_login — recency matters most.
- Low multicollinearity among features confirms all variables contribute independent predictive value.

Step 3 | Data Cleaning & Outlier Handling

01

Drop Identifier Columns

Removed CustomerID, TransactionID, InteractionID — non-predictive unique keys

Dropped

02

Missing Values Check

`df.isnull().sum()` → 0 missing values across all 5,204 rows and 17 columns

0 Missing

03

Outlier Capping (IQR)

Applied IQR method to Age, AmountSpent, LoginFrequency → values clipped to $Q1 - 1.5 \cdot IQR$ / $Q3 + 1.5 \cdot IQR$ bounds

Capped

04

Label Encoding

Encoded: Gender, MaritalStatus, IncomeLevel, InteractionType, ResolutionStatus, ServiceUsage using LabelEncoder

Encoded

05

Standard Scaling

StandardScaler on: Age, AmountSpent, LoginFrequency, day_since_last_login, TotalInteraction → mean=0, std=1

Scaled

Step 3 | Final Preprocessed Dataset — Ready for Modelling

5,204 rows × 17 columns · All numeric · Zero nulls · Scaled & encoded · Model ready

Feature	Type	Transformation	Role
Age	Numeric	StandardScaler	Customer age
Gender	Encoded	LabelEncoder 0/1	M=1, F=0
MaritalStatus	Encoded	LabelEncoder 0-3	4 categories
IncomeLevel	Encoded	LabelEncoder 0-2	3 income tiers
AmountSpent	Numeric	StandardScaler	Transaction value
ProductCategory	String	As-is	Purchase category
InteractionType	Encoded	LabelEncoder	Support type
ResolutionStatus	Encoded	LabelEncoder 0/1	Resolved flag
LoginFrequency	Numeric	StandardScaler	Digital engagement
ServiceUsage	Encoded	LabelEncoder	Channel (app/web)
day_since_last_login	Numeric	StandardScaler	Engineered – recency
TransactionMonth	Numeric	As-is	Engineered – seasonality
TotalInteraction	Numeric	StandardScaler	Engineered – support volume
ChurnStatus	Binary	Target variable	0 = Retained, 1 = Churned

Note: TransactionDate, InteractionDate, LastLoginDate still present — drop or convert before model training.

Key Findings & Next Steps



Data Quality

Zero missing values, 5 datasets successfully merged. Solid foundation for ML.



Churn Signal

20.6% churn rate — sufficient positive class for balanced model training.



Best Predictors

LoginFrequency and day_since_last_login show strongest correlation with ChurnStatus.



Next: Modelling

Train Logistic Regression, Random Forest and XGBoost. Evaluate with F1, AUC-ROC.